

Warehouse Operations Optimization

Order Batching and Picking Routing

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1 Introduction

This project is dedicated to optimizing warehouse operations, with a focus on **order batching** and **picking processes**. Efficient batching and picking are critical for reducing operational costs, minimizing travel distances, and improving overall order fulfillment performance.

This project was proposed by **Find&Order**, a company specializing in warehouse and logistics solutions.

In addition to exact mathematical models, the project explores **heuristic approaches** to generate high-quality solutions within reasonable computation times. These heuristics are particularly relevant in real-world warehouses, where the combinatorial complexity of batching and routing problems often makes exact optimization impractical.

The objective of this document is to present both the **mathematical formulations** of the batching and picking problems and the **heuristic methods** developed to address them. This provides a comprehensive view of the strategies used to improve warehouse efficiency.

2 Problem Description

This project addresses the optimization of warehouse order preparation operations by jointly considering *order batching* and *picking routing* problems.

2.1 Warehouse Environment

The warehouse is modeled as a set of storage positions where the products are located. The layout of the warehouse is represented through:

- A set of storage locations.
- An adjacency matrix describing the travel distance between every pair of locations.
- A predefined starting location and ending location for each picking tour.

A set of customer orders is provided. Each order is associated with:

- A unique identifier.
- A volume representing its physical size.
- A list of storage locations corresponding to the products that must be collected.

In this project, each order is assumed to be associated with a dedicated container, referred to as a *support*. This assumption reflects the operational framework considered and simplifies traceability and batching decisions.

During warehouse operations, a picker can transport several supports simultaneously during a single picking tour. The set of supports collected during the same picking tour is referred to as a *batch*.

2.2 Decision Problem

The decision problem consists of determining:

1. How to assign orders to batches.
2. The sequence in which storage locations are visited for each batch.

The objective is to minimize the total travel distance required to collect all products across all batches.

2.3 Batching Constraints

The batching decisions must satisfy the following operational constraints:

- Each order must be assigned to exactly one batch.
- The number of supports assigned to a batch cannot exceed a predefined maximum capacity.
- The total volume of supports assigned to a batch cannot exceed a predefined maximum volume.

2.4 Picking Constraints

For each batch, the picking route must satisfy the following requirements:

- Every storage location corresponding to the products of the batch must be visited.
- The picking route must start at the warehouse depot.
- The picking route must end at the designated exit location.

2.5 Objective Function

The cost of a solution is defined as the sum of travel distances required to complete all batch picking routes. The performance of a solution is evaluated by comparing it to a baseline solution in which each order is processed independently. In this baseline approach, each support is assigned to a single batch, and the picking route follows a predefined serpentine routing strategy, commonly referred to as an S-shape path, which corresponds to the default routing rule used in many warehouse management systems.

2.6 Computational Complexity

The problem addressed in this project combines two well-known combinatorial optimization problems:

- The Order Batching Problem, which is closely related to bin packing and clustering problems.
- The Picking Routing Problem, which can be formulated as a variation of the Traveling Salesman Problem (TSP).

Both subproblems are known to be NP-hard. Consequently, the integrated batching and routing problem exhibits exponential growth in the number of feasible solutions as the number of orders increases.

Due to this computational complexity, solving large-scale instances optimally becomes impractical within reasonable computation times. Therefore, heuristic and metaheuristic approaches are required to generate high-quality solutions efficiently in realistic industrial environments.

3 Mathematical Models

This section presents the mathematical formulations used to model the order batching and picking routing problems described in Section 2. The models are developed following a sequential optimization framework. First, orders are grouped into batches. Then, a picking route is computed for each batch.

In addition, a global (main) model is introduced to jointly solve both the batching and routing decisions within a single integrated optimization framework.

3.1 Order Batching Model

3.1.1 Context

The batching model aims to assign customer orders to batches while satisfying operational constraints such as batch capacity and volume limitations. The objective is to group orders that are likely to be collected together efficiently, thereby reducing the travel distance required during picking operations.

Since the exact picking route is not known at the batching stage, an order similarity measure based on shared picking locations is used as a proxy to estimate the potential travel distance reduction obtained by grouping orders.

3.1.2 Sets

- O : set of customer orders, indexed by o .
- B : set of potential batches, indexed by b .

3.1.3 Parameters

- v_o : volume associated with order o .
- C : maximum volume capacity of a batch.
- N : maximum number of supports allowed in a batch.
- $a_{oo'}$: number of shared picking locations between orders o and o' .

3.1.4 Decision Variables

- $y_{ob} \in \{0, 1\}$: equals 1 if order o is assigned to batch b , 0 otherwise.
- $z_{oo'b} \in \{0, 1\}$: equals 1 if both orders o and o' are assigned to the same batch b , 0 otherwise.

The variable $z_{oo'b}$ is introduced to linearize the quadratic term $y_{ob} \cdot y_{o'b}$ appearing in the objective function. It ensures that $z_{oo'b} = 1$ if and only if both orders o and o' are assigned to batch b .

3.1.5 Objective Function

The batching objective is to maximize the similarity between orders assigned to the same batch:

$$\max \sum_{b \in B} \sum_{\substack{o, o' \in O \\ o < o'}} a_{oo'} z_{oo'b} \quad (1)$$

This objective promotes grouping orders sharing common storage locations, which is expected to reduce picking travel distance.

3.1.6 Constraints

1. Each order must be assigned to exactly one batch

$$\sum_{b \in B} y_{ob} = 1 \quad \forall o \in O \quad (2)$$

2. Batch volume constraint

$$\sum_{o \in O} v_o y_{ob} \leq C \quad \forall b \in B \quad (3)$$

3. Batch size constraint

$$\sum_{o \in O} y_{ob} \leq N \quad \forall b \in B \quad (4)$$

4. Linearization constraints

$$z_{oo'b} \leq y_{ob} \quad \forall o, o' \in O, o < o', \forall b \in B \quad (5)$$

$$z_{oo'b} \leq y_{o'b} \quad \forall o, o' \in O, o < o', \forall b \in B \quad (6)$$

$$z_{oo'b} \geq y_{ob} + y_{o'b} - 1 \quad \forall o, o' \in O, o < o', \forall b \in B \quad (7)$$

5. Domain of the variables

$$y_{ob} \in \{0, 1\}, \quad \forall o \in O, \forall b \in B \quad (8)$$

$$z_{oo'b} \in \{0, 1\}, \quad \forall o, o' \in O, o < o', \forall b \in B \quad (9)$$

3.2 Order Picking Model

3.2.1 Context

The picking model aims to determine efficient routes for each picker within the warehouse while satisfying operational constraints. The objective is to minimize the total travel distance by optimizing the sequence in which picking locations are visited. Based on the warehouse layout and adjacency structure, the model computes optimal routes that improve efficiency and reduce overall picking time.

3.2.2 Sets

- P : set of pickers.
- L_p : set of locations to be visited by picker $p \in P$.

3.2.3 Parameters

- A_{ij} : traveling distance from location i to location j .
- s_p : starting location of picker p (first element of L_p).
- t_p : ending location of picker p (last element of L_p).

3.2.4 Decision Variables

- $x_{ijp} \in \{0, 1\}$: equals 1 if picker p travels from location i to j , 0 otherwise.
- $u_{ip} \in \mathbb{Z}$: auxiliary variable used for sub-tour elimination (MTZ formulation).

3.2.5 Objective Function

$$\min \sum_{p \in P} \sum_{i \in L_p} \sum_{\substack{j \in L_p \\ j \neq i}} A_{ij} x_{ijp} \quad (10)$$

3.2.6 Constraints

1. Each location (except the start) has exactly one predecessor

$$\sum_{\substack{i \in L_p \\ i \neq j}} x_{ijp} = 1 \quad \forall p \in P, \forall j \in L_p \setminus \{s_p\} \quad (11)$$

2. Each location (except the end) has exactly one successor

$$\sum_{\substack{j \in L_p \\ j \neq i}} x_{ijp} = 1 \quad \forall p \in P, \forall i \in L_p \setminus \{t_p\} \quad (12)$$

3. No outgoing arc from the end location

$$x_{t_pjp} = 0 \quad \forall p \in P, \forall j \in L_p, j \neq t_p \quad (13)$$

4. Sub-tour elimination constraints (MTZ)

$$u_{ip} - u_{jp} + |L_p| \cdot x_{ijp} \leq |L_p| - 1 \quad \forall p \in P, \forall i, j \in L_p, i \neq j, i \neq t_p, j \neq s_p \quad (14)$$

5. Fixing MTZ variables

$$u_{s_pp} = 0 \quad \forall p \in P \quad (15)$$

$$u_{t_pp} = |L_p| - 1 \quad \forall p \in P \quad (16)$$

6. Domain of the variables

$$x_{ijp} \in \{0, 1\} \quad \forall p \in P, \forall i, j \in L_p, i \neq j \quad (17)$$

$$u_{ip} \in \{0, \dots, |L_p| - 1\} \quad \forall p \in P, \forall i \in L_p \quad (18)$$

3.3 Integrated Order Batching and Routing Model

3.3.1 Sets

- O : set of orders.
- P : set of pickers.
- L : set of locations to be visited.

3.3.2 Parameters

- A_{ij} : travel distance from location i to j .
- v_o : volume associated with order o .
- a_{io} : equals 1 if order o requires location i , 0 otherwise.
- C : maximum volume capacity per batch.
- N : maximum number of orders per batch.
- M : a sufficiently large constant (e.g., $|L|$).
- s : start location in L .
- t : end location in L .

3.3.3 Decision Variables

- $y_{bo} \in \{0, 1\}$: equals 1 if order o is assigned to batch b .
- $h_{pi} \in \{0, 1\}$: equals 1 if picker p visits location i .
- $x_{ijp} \in \{0, 1\}$: equals 1 if picker p travels from i to j .
- $u_{ip} \in \mathbb{N}$: MTZ variable for sub-tour elimination.

3.3.4 Objective Function

$$\min \sum_{p \in P} \sum_{i \in L} \sum_{\substack{j \in L \\ j \neq i}} A_{ij} x_{ijp} \quad (19)$$

3.3.5 Constraints

1. Each order is assigned to exactly one batch

$$\sum_{b \in P} y_{bo} = 1 \quad \forall o \in O \quad (20)$$

2. Maximum number of orders per batch

$$\sum_{o \in O} y_{bo} \leq N \quad \forall b \in P \quad (21)$$

3. Capacity constraint (volume)

$$\sum_{o \in O} v_o y_{bo} \leq C \quad \forall b \in P \quad (22)$$

4. Linking orders and visited locations

$$\sum_{o \in O} a_{io} y_{bo} \geq h_{bi} \quad \forall b \in P, \forall i \in L \quad (23)$$

$$y_{bo} \cdot a_{io} \leq h_{bi} \quad \forall b \in P, \forall i \in L, \forall o \in O \quad (24)$$

5. Flow conservation (incoming arcs)

$$\sum_{\substack{i \in L \\ i \neq j}} x_{ijp} = h_{pj} \quad \forall p \in P, \forall j \in L \setminus \{s\} \quad (25)$$

6. Flow conservation (outgoing arcs)

$$\sum_{\substack{j \in L \\ j \neq i}} x_{ijp} = h_{pi} \quad \forall p \in P, \forall i \in L \setminus \{t\} \quad (26)$$

7. No outgoing arc from the end location

$$x_{tjp} = 0 \quad \forall p \in P, \forall j \in L, j \neq t \quad (27)$$

8. Sub-tour elimination (MTZ)

$$u_{ip} - u_{jp} + M \cdot x_{ijp} \leq M - 1 \quad \forall p \in P, \forall i, j \in L, i \neq j, i \neq t, j \neq s \quad (28)$$

9. MTZ variable fixing

$$u_{sp} = 0 \quad \forall p \in P \quad (29)$$

$$u_{tp} \geq M - 1 \quad \forall p \in P \quad (30)$$

10. Variable Domains

$$y_{bo} \in \{0, 1\} \quad \forall b \in P, \forall o \in O \quad (31)$$

$$h_{pi} \in \{0, 1\} \quad \forall p \in P, \forall i \in L \quad (32)$$

$$x_{ijp} \in \{0, 1\} \quad \forall p \in P, \forall i, j \in L, i \neq j \quad (33)$$

$$u_{ip} \in \{0, \dots, M - 1\} \quad \forall p \in P, \forall i \in L \quad (34)$$

4 Heuristic Methods

Due to the NP-hard nature of the integrated Order Batching and Picker Routing Problem, finding an optimal solution through exact methods within operational time constraints is often computationally prohibitive. Consequently, a decoupled heuristic framework is adopted. This approach decomposes the problem into two sequential decision-making phases: first, grouping orders into capacity-constrained batches, and second, determining the optimal traversal sequence for each picker. This modular approach allows for high-quality, scalable solutions that remain adaptable to real-time warehouse constraints.

4.1 Order Batching Heuristics

The primary objective of order batching is to consolidate multiple customer orders into a single picking tour to minimize the total travel distance. By grouping orders that share common storage locations or are located in close proximity, the warehouse can significantly reduce "deadhead" travel time (empty travel between picks). The following heuristics employ a "seed-and-grow" constructive strategy to build initial batches, followed by iterative local search improvements designed to maximize the spatial affinity between orders within each batch.

4.1.1 Greedy Order Batching

The greedy batching algorithm builds batches iteratively using a seed-and-grow strategy.

Seed selection. At each iteration, the unassigned order sharing the most locations with remaining unassigned orders is selected as the seed of a new batch:

$$o_{\text{seed}}^* = \arg \max_{o \in U} \sum_{\substack{o' \in U \\ o' \neq o}} a_{oo'}$$

where U denotes the set of unassigned orders and $a_{oo'}$ is the number of shared locations between orders o and o' .

Batch growth. Once the seed is fixed, the algorithm greedily adds the most compatible unassigned order to the batch:

$$o^* = \arg \max_{o \in U} \sum_{o' \in \text{batch}} a_{oo'}$$

subject to the capacity constraints:

$$|\text{batch}| + 1 \leq N \quad \text{and} \quad \sum_{o' \in \text{batch}} v_{o'} + v_o \leq C$$

Stopping criterion. The batch growth stops when one of the following conditions is met:

- No unassigned order satisfies the capacity constraints:

$$|\text{batch}| + 1 > N \quad \text{or} \quad \sum_{o' \in \text{batch}} v_{o'} + v_o > C \quad \forall o \in U$$

- No candidate improves the score:

$$\sum_{o' \in \text{batch}} a_{oo'} = 0 \quad \forall o \in U$$

A new batch is then initialized until all orders are assigned, i.e. $U = \emptyset$.

4.1.2 Local Search by Swap

Starting from an initial batching solution (e.g., obtained by the greedy algorithm), this metaheuristic iteratively attempts to improve the solution by swapping two orders between two randomly selected batches.

At each iteration:

1. Two distinct batches b_1 and b_2 are randomly selected.
2. One order $o_1 \in b_1$ and one order $o_2 \in b_2$ are randomly selected.
3. The swap is performed if and only if the volume constraints are satisfied for both batches after the exchange.
4. The new solution is accepted only if it strictly improves the total score:

$$\text{score}(\sigma) = \sum_b \sum_{\substack{o, o' \in b \\ o < o'}} a_{oo'}$$

The search runs until a predefined time limit is reached.

4.1.3 Local Search by Move

Similar to the swap heuristic, this metaheuristic improves an initial solution by moving a single order from one batch to another.

At each iteration:

1. A source batch b_1 is randomly selected.
2. An order $o \in b_1$ is randomly selected.
3. A target batch $b_2 \neq b_1$ is randomly selected.
4. The move is performed if the volume constraint is satisfied for b_2 after adding o .
5. The new solution is accepted only if it strictly improves the total score.

The move heuristic is complementary to the swap heuristic: it allows rebalancing batch sizes, which the swap operation cannot achieve since it preserves batch sizes. The search also runs until a predefined time limit is reached.

4.2 Picking Routing Heuristics

Once the batches are defined, the routing phase determines the exact sequence of locations the picker must visit. This is essentially a Traveling Salesman Problem (TSP) variation where the goal is to find the shortest Hamiltonian path starting and ending at the depot. Since pickers often face high-density picking lists, the routing strategy must balance computational speed with path efficiency. We first implement a greedy constructive heuristic to establish a baseline route, which is then refined using metaheuristic local search techniques to eliminate spatial inefficiencies.

4.2.1 Nearest Neighbor Routing

Once orders are assigned to batches, the routing problem determines the sequence in which the picker visits the required locations. Given the complexity of the Traveling Salesman Problem (TSP) structure within each batch, a *Nearest Neighbor* (NN) constructive heuristic is implemented to provide rapid, high-quality routes.

Algorithm Logic. For each batch b , let $L_b = \{l_0, l_1, \dots, l_n, l_e\}$ be the set of locations, where l_0 is the starting depot and l_e is the ending depot. The algorithm proceeds as follows:

1. Initialize the current location at the starting depot: $c = l_0$.
2. Define the set of locations to visit: $V = \{l_1, \dots, l_n\}$.
3. While $V \neq \emptyset$:
 - Find the nearest location $l^* \in V$ from the current position c :

$$l^* = \arg \min_{l \in V} \text{dist}(c, l)$$

- Add the arc (c, l^*) to the route and update the total distance.
 - Set $c = l^*$ and remove l^* from V .
4. Add the final arc (c, l_e) back to the ending depot.

Performance Metrics. To evaluate the efficiency of the resulting routes, two density metrics (ρ) are defined based on the relationship between the number of items collected and the distance traveled:

- **Local Batch Density:** Measures the efficiency of an individual picker p . If K_p is the number of picks in batch p and D_p is the distance traveled in meters:

$$\rho_p^{\text{local}} = \frac{K_p}{D_p}$$

The dispersion of these values is analyzed using $\rho^{\min} = \min(\rho^{\text{local}})$ and $\rho^{\max} = \max(\rho^{\text{local}})$.

- **Global System Density:** Measures the overall productivity of the warehouse operations:

$$\rho^{\text{global}} = \frac{\sum_p K_p}{\sum_p D_p}$$

These metrics allow for a dual-perspective analysis: ensuring both economic profitability for the warehouse and workload consistency for the pickers.

4.2.2 Local Search for Routing: 2-Opt

To further improve the routes generated by the Nearest Neighbor heuristic, a *2-Opt* local search is applied. This metaheuristic systematically tests the removal of two non-adjacent arcs and replaces them with two new arcs to reconnect the route in a shorter sequence.

Mechanism. Let a route be defined as a sequence of nodes. The algorithm selects two edges from the current tour:

- Edge 1: (i, j) , where j is the immediate successor of i .
- Edge 2: (k, l) , where l is the immediate successor of k .

A 2-opt swap consists of deleting these two edges and creating two new edges: (i, k) and (j, l) . To maintain a single continuous and valid tour, the entire segment of the route originally located between j and k is reversed. This ensures that every pick location is still visited exactly once.

Improvement Criterion. A swap is accepted if and only if it results in a strict reduction of the total travel distance. Specifically, an improvement is found if:

$$\text{dist}(i, k) + \text{dist}(j, l) < \text{dist}(i, j) + \text{dist}(k, l)$$

where $\text{dist}(u, v)$ represents the travel distance between nodes u and v retrieved from the adjacency matrix.

This process is repeated iteratively for all possible pairs of edges until no further swaps can reduce the total distance. By eliminating crossings (the "X" patterns), the 2-opt heuristic maximizes the *Global Density* of the picking operation by reducing the denominator (total distance) while keeping the numerator (number of picks) constant.

5 Experimentations

This section presents the computational experiments conducted to evaluate the performance of the integrated batching and routing heuristics. We compare the heuristic pipelines against the theoretical lower bounds provided by the mathematical model relaxations.

5.1 Experimental Instances

The performance of the algorithms is tested using two distinct warehouse datasets, representing different scales of operation and layout complexities:

- **Warehouse A (Wh. A):** A small-to-medium scale facility featuring 277 picking locations and 20 to 30 **orders**. This site serves as the baseline to verify the fundamental logic of the heuristic pipeline. Operational constraints are set to a maximum of 2 to 4 **orders** per batch with a total volume capacity of 70 units.
- **Warehouse B (Wh. B):** A large-scale industrial facility with 577 picking locations and 60 to 90 **orders**. This environment acts as a stress test for the integrated approach. The combinatorial search space is significantly expanded by a higher batch density of up to 6 **orders** per picker and a volume capacity of 70 units, testing the heuristics' ability to maintain efficiency where mathematical models struggle.

5.2 Performance Metrics

The effectiveness of the proposed methods is evaluated through two distinct sets of metrics, corresponding to the global benchmark of the routing refinement and the comparative analysis of the local search operators.

5.2.1 Case 1: Global Routing Benchmark and Batch Density

For the initial comparison between the mathematical baseline and the routing heuristics (Table 1), the following metrics are utilized:

- **Distance (m):** The total travel distance across all batches. For the **Picking Model (Relax)**, this represents a **mathematical lower bound** obtained by relaxing variables. It serves as a benchmark for the minimum possible distance, against which the efficiency of the heuristic paths is measured.
- **Computational Time (s):** The total duration required to generate a solution, measuring the execution of the mathematical solver for the **Picking Model** and the **heuristics**.
- **Gap (%):** The percentage deviation of the heuristic distance from the relaxed model lower bound, calculated as:

$$\text{Gap} = \frac{\text{Distance}_{\text{Heuristic}} - \text{Distance}_{\text{Relax}}}{\text{Distance}_{\text{Relax}}} \times 100$$

- **Density Metrics (ρ):** A set of indicators used to quantify the concentration of picking locations relative to the travel distance within the adjacency matrix. This metric serves as a proxy for picking efficiency:
 - **Global Density:** Calculated as the total number of picks in an instance divided by the total travel distance across all batches ($\rho_{global} = \frac{\sum \text{Picks}}{\sum \text{Distance}}$). A higher value indicates a reduction in "deadhead" travel between picking nodes.
 - **Min/Max Density:** These represent the density values of the single most dispersed and most clustered batches within an instance, respectively ($\rho = \frac{n_{picks}}{\text{dist}_{batch}}$). They highlight the variance in batch quality and the algorithm's ability to create geographically cohesive clusters.

5.2.2 Case 2: Full Heuristic Pipeline and Operator Synergy

This experimental phase (Table 2) evaluates the performance of the complete integrated heuristic suite. We compare the baseline **Main Model (Relax)** against various heuristic configurations, including the initial **Greedy + NN** and enhanced pipelines incorporating batching operators (**Swap/Move**) and routing optimization (**2-Opt**). The metrics utilized are:

- **Distance (m):** The total travel distance across all batches. For the **Main Model (Relax)**, this represents the theoretical lower bound, while for the heuristics, it is the actual path length calculated via the adjacency matrix.
- **Time (s):** The total computational duration required to generate the solution. This includes the solver's search time for the mathematical model and the execution of the entire heuristic pipeline (batching and routing) for the algorithms.
- **Gap (%):** The percentage deviation of the heuristic distance from the **Main Model (Relax)** baseline, calculated as:

$$\text{Gap} = \frac{\text{Distance}_{\text{Heuristic}} - \text{Distance}_{\text{Relax}}}{\text{Distance}_{\text{Relax}}} \times 100$$

5.3 Analysis of Results

The results from Table 1 indicate that the **NN + 2-Opt** pipeline significantly outperforms the standard **NN** approach across all instances. By applying local search to re-sequence the pick locations, the **2-Opt** algorithm consistently reduces the distance gap by approximately **8–10%**. A critical observation is the positive correlation between routing refinement and the **Global Density (ρ)**. As the total travel distance decreases, the density increases, signifying a more compact utilization of the warehouse aisles and a reduction in "deadhead" travel. The stability of the **Gap** (averaging 30% across both warehouses) suggests that the routing heuristic is highly robust and maintains consistent performance regardless of the warehouse scale or the number of batches.

The comparison in Table 2 highlights a definitive boundary for mathematical solvers in industrial applications. While the **Main Model (Relax)** provides a deep theoretical lower

bound, its computational cost becomes prohibitive in **Wh. B**, requiring over **5,100 seconds** per instance. In contrast, the integrated heuristic suite—combining **Swap** or **Move** operators with **2-Opt**—reaches a solution in approximately **0.03 seconds**.

Although the heuristics exhibit a high gap relative to the heavily relaxed model, they demonstrate superior practical utility. The **Swap-based** and **Move-based** pipelines show comparable performance, with the **Move** operator slightly outperforming **Swap** in instances with higher order counts (e.g., Wh. B 05-24).

6 Conclusion and Future Perspectives

This study evaluated a tiered heuristic pipeline designed to address the joint order batching and picker routing problem within a graph-based warehouse model. The experimental results confirm that while exact mathematical formulations provide a rigorous theoretical baseline, their computational requirements grow prohibitively as the problem dimensions increase. By implementing a sequential optimization strategy—combining **Swap** and **Move** operators for batch redistribution with **2-Opt** for routing refinement—the proposed approach delivers solutions several orders of magnitude faster than the **Main Model (Relax)**. In large-scale environments, where the relaxed model requires over an hour to converge, the heuristic pipeline provides feasible results in milliseconds. This performance gap demonstrates that for industrial logistics, the trade-off in optimality is a necessary and justified requirement for real-time decision-making.

While the current heuristic suite provides a robust framework for static batching and routing, several avenues for future evolution remain:

- **Dynamic Order Integration:** Adapting the **Move** operator to handle real-time order arrivals, allowing for the insertion of new picks into active routes without triggering a full re-batching process.
- **Congestion Awareness:** Modifying the adjacency matrix to include dynamic edge weights that penalize high picker density. This would allow the model to optimize for total throughput by avoiding physical bottlenecks in high-traffic aisles.

Table 1: Full Benchmark Comparison: Picking Model Relaxation vs. NN vs. NN + 2-Opt

Site	Instance	Picking Method	Num.	Distance	Time	Gap	Density Metrics (ρ)		
			Batches	(m)	(s)	(%)	Global	Min	Max
Wh. A	05-22	Picking Model (Relax)	12	259	0.48	–	–	–	–
		NN	12	427	0.00	39.31	0.83	0.54	1.04
		NN + 2-Opt	12	376	0.01	31.03	0.94	0.65	1.10
	05-23	Picking Model (Relax)	11	260	0.40	–	–	–	–
		NN	11	416	0.00	37.51	0.79	0.65	0.97
		NN + 2-Opt	11	359	0.01	27.55	0.91	0.75	1.11
	05-24	Picking Model (Relax)	11	245	0.54	–	–	–	–
		NN	11	396	0.00	38.18	0.85	0.65	0.95
		NN + 2-Opt	11	354	0.01	30.92	0.95	0.75	1.13
	05-25	Picking Model (Relax)	14	324	0.63	–	–	–	–
		NN	14	515	0.00	37.01	0.84	0.57	1.05
		NN + 2-Opt	14	454	0.01	28.63	0.95	0.65	1.24
	05-26	Picking Model (Relax)	23	472	0.65	–	–	–	–
		NN	23	807	0.00	41.44	0.72	0.44	0.96
		NN + 2-Opt	23	684	0.01	30.92	0.84	0.51	1.15
	05-27	Picking Model (Relax)	18	412	0.60	–	–	–	–
		NN	18	665	0.00	38.02	0.77	0.37	1.09
		NN + 2-Opt	18	576	0.01	28.46	0.89	0.42	1.17
Wh. B	05-22	Picking Model (Relax)	38	869	1.21	–	–	–	–
		NN	38	1439	0.00	39.59	0.81	0.57	1.04
		NN + 2-Opt	38	1255	0.02	30.77	0.92	0.62	1.10
	05-23	Picking Model (Relax)	31	713	0.95	–	–	–	–
		NN	31	1169	0.00	38.97	0.82	0.30	0.99
		NN + 2-Opt	31	1027	0.02	30.54	0.93	0.35	1.14
	05-24	Picking Model (Relax)	39	906	1.34	–	–	–	–
		NN	39	1483	0.00	38.94	0.85	0.55	1.21
		NN + 2-Opt	39	1284	0.02	29.46	0.98	0.64	1.30
	05-25	Picking Model (Relax)	34	844	1.27	–	–	–	–
		NN	34	1381	0.00	38.88	0.84	0.62	1.12
		NN + 2-Opt	34	1203	0.03	29.83	0.97	0.71	1.30
	05-26	Picking Model (Relax)	60	1377	1.86	–	–	–	–
		NN	60	2232	0.01	38.32	0.80	0.44	1.32
		NN + 2-Opt	60	1929	0.03	28.64	0.93	0.51	1.48
	05-27	Picking Model (Relax)	40	914	1.22	–	–	–	–
		NN	40	1464	0.00	37.58	0.81	0.37	1.35
		NN + 2-Opt	40	1291	0.02	29.23	0.92	0.42	1.39

Note: All heuristic paths start from the **Greedy** batching solution. **Picking Model (Relax)** represents the theoretical lower bound. **NN** is a greedy Nearest Neighbor routing; **NN + 2-Opt** applies local search to the NN path to untangle and shorten the total distance. Gap (%) is measured against the Relaxed Model baseline. Bold values indicate the best heuristic performance.

Table 2: Benchmark Comparison: Main Model Relaxation and Heuristic Combinations

Site	Instance	Method	Parameters		Pickers	Distance (m)	Time (s)	Gap (%)
			Orders	Loc.				
Wh. A	05-22	Main Model (Relax)	20	277	–	54	86	–
		Greedy + NN	20	277	12	427	0.00	87.38
		Greedy+Swap+NN+2-opt	20	277	12	374	0.01	85.60
		Greedy+Move+NN+2-opt	20	277	12	375	0.01	85.66
	05-23	Main Model (Relax)	20	277	–	55	91	–
		Greedy + NN	20	277	11	416	0.00	86.67
		Greedy+Swap+NN+2-opt	20	277	11	357	0.01	84.46
		Greedy+Move+NN+2-opt	20	277	11	359	0.01	84.54
	05-24	Main Model (Relax)	20	277	–	50	65	–
		Greedy + NN	20	277	11	396	0.00	87.35
		Greedy+Swap+NN+2-opt	20	277	11	357	0.01	85.96
		Greedy+Move+NN+2-opt	20	277	11	354	0.01	85.86
	05-25	Main Model (Relax)	25	277	–	57	191	–
		Greedy + NN	25	277	14	515	0.00	89.02
		Greedy+Swap+NN+2-opt	25	277	14	457	0.01	87.61
		Greedy+Move+NN+2-opt	25	277	14	454	0.01	87.55
	05-26	Main Model (Relax)	30	277	–	61	1,023	–
		Greedy + NN	30	277	23	806	0.00	92.47
		Greedy+Swap+NN+2-opt	30	277	23	669	0.01	90.93
		Greedy+Move+NN+2-opt	30	277	23	684	0.01	91.12
	05-27	Main Model (Relax)	30	277	–	57	842	–
		Greedy + NN	30	277	18	665	0.00	91.40
		Greedy+Swap+NN+2-opt	30	277	18	571	0.01	89.99
		Greedy+Move+NN+2-opt	30	277	18	576	0.01	90.07
Wh. B	05-22	Main Model (Relax)	71	577	–	87	5,034	–
		Greedy + NN	71	577	38	1,438	0.00	93.91
		Greedy+Swap+NN+2-opt	71	577	38	1,254	0.02	93.02
		Greedy+Move+NN+2-opt	71	577	38	1,257	0.02	93.04
	05-23	Main Model (Relax)	60	577	–	67	5,124	–
		Greedy + NN	60	577	31	1,168	0.00	94.19
		Greedy+Swap+NN+2-opt	60	577	31	1,027	0.01	93.39
		Greedy+Move+NN+2-opt	60	577	31	1,026	0.01	93.39
	05-24	Main Model (Relax)	64	577	–	99	5,082	–
		Greedy + NN	64	577	39	1,483	0.00	93.31
		Greedy+Swap+NN+2-opt	64	577	39	1,279	0.02	92.25
		Greedy+Move+NN+2-opt	64	577	39	1,274	0.02	92.22
	05-25	Main Model (Relax)	60	577	–	94	5,073	–
		Greedy + NN	60	577	34	1,381	0.00	93.15
		Greedy+Swap+NN+2-opt	60	577	34	1,196	0.02	92.10
		Greedy+Move+NN+2-opt	60	577	34	1,200	0.02	92.13
	05-26	Main Model (Relax)	90	577	–	204	5,131	–
		Greedy + NN	90	577	60	2,232	0.01	90.84
		Greedy+Swap+NN+2-opt	90	577	60	1,925	0.03	89.38
		Greedy+Move+NN+2-opt	90	577	60	1,925	0.03	89.38
	05-27	Main Model (Relax)	70	577	–	76	5,102	–
		Greedy + NN	70	577	40	1,463	0.00	94.74
		Greedy+Swap+NN+2-opt	70	577	40	1,288	0.02	94.03
		Greedy+Move+NN+2-opt	70	577	40	1,284	0.02	94.01

Note: All heuristic pipelines originate from the **Greedy** solution. **Main Model (Relax)** serves as the theoretical lower bound for distance. **Swap** and **Move** are batching operators used to redistribute **orders** between pickers, while **NN + 2-opt** optimizes the final routing path. Gap (%) represents the distance increase relative to the Relaxed Model. Bold values denote the best-performing heuristic combination.